

BaseServlet.java

```
package com.harisudhan.harisudhanmart.controller;

import com.harisudhan.harisudhanmart.dao.DAOFactory;
import com.harisudhan.harisudhanmart.dto.ApiResponse;
import com.harisudhan.harisudhanmart.listener.AppContextListener;
import com.harisudhan.harisudhanmart.util.JsonUtil;
import java.io.IOException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

/**
 * Shared helpers for all controller servlets. Servlets stay thin (HTTP orchestration only,
 * per Section 12 SOLID rule) – validation and business logic live in the service layer.
 */
public abstract class BaseServlet extends HttpServlet {

    protected DAOFactory daoFactory() {
        return new DAOFactory(AppContextListener.getDataSource());
    }

    protected void writeJson(HttpServletResponse resp, int status, Object data) throws IOException {
        resp.setStatus(status);
        resp.setContentType("application/json");
        resp.getWriter().write(JsonUtil.gson().toJson(ApiResponse.ok(data)));
    }

    protected void writeError(HttpServletResponse resp, int status, String code, String message) throws IOException {
        resp.setStatus(status);
        resp.setContentType("application/json");
        resp.getWriter().write(JsonUtil.gson().toJson(ApiResponse.fail(code, message)));
    }

    protected Long currentUserId(HttpSession session) {
        return session == null ? null : (Long) session.getAttribute("userId");
    }

    protected String currentRole(HttpSession session) {
        return session == null ? null : (String) session.getAttribute("role");
    }
}
```